

Computer Networks

Protocol Architecture and TCP/IP

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Protocol “layers” and reference models

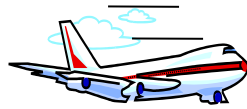
Networks are complex, with many “pieces”:

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

Question: is there any hope of *organizing* structure of network?

Solution:
Layering

Example: organization of air travel



— *end-to-end transfer of person plus baggage*

ticket (purchase)

baggage (check)

gates (load)

runway takeoff

airplane routing

ticket (complain)

baggage (claim)

gates (unload)

runway landing

airplane routing

airplane routing

Example: organization of air travel

ticket (purchase)	<i>ticketing service</i>	ticket (complain)
baggage (check)	<i>baggage service</i>	baggage (claim)
gates (load)	<i>gate service</i>	gates (unload)
runway takeoff	<i>runway service</i>	runway landing
airplane routing	<i>routing service</i>	airplane routing

layers: each layer implements a service

Why layering?

Approach to designing/discussing complex systems:

- explicit structure allows identification, relationship of system's pieces
 - modularization eases maintenance, updating of system
 - change in layer's service *implementation*: transparent to rest of system
 - e.g., change in gate procedure doesn't affect rest of system
-

Protocol Models

□ OSI model: Open System Interconnection

- Developed by International Standard Organization (ISO)
- 7 layers

□ TCP/IP model:

- Developed by DARPA for first generation packet switched networks (ARPANET)
 - 5 Layers
 - Used by global Internet
-

TCP/IP Layers

**User
Processes**



**Operating
Systems**



**Network Interface
Cards**



application

transport

network

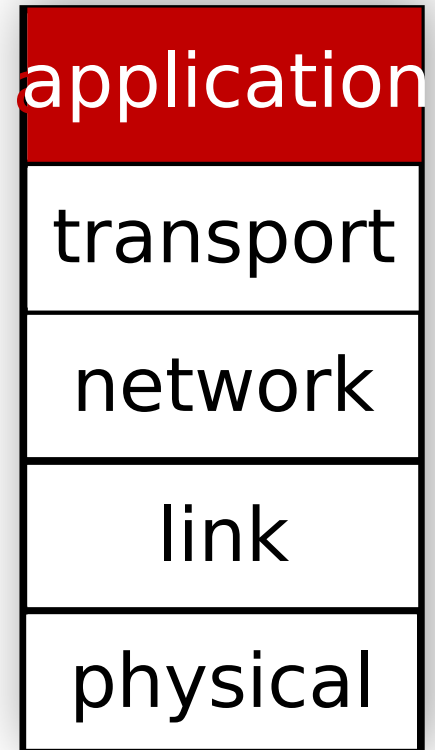
link

physical

Application Layer

□ Application layer:

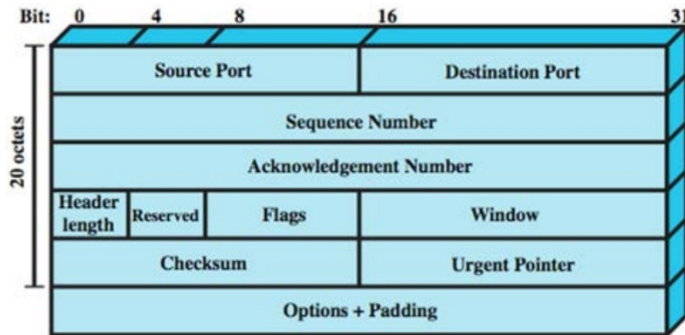
- Support user application
 - Web browsing: HTTP
 - File Transfer: FTP
 - Email: SMTP



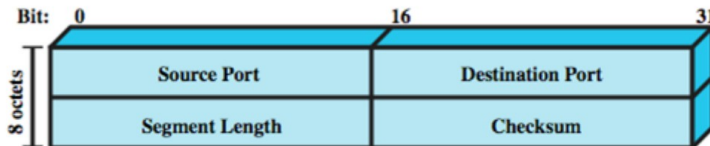
Transport Layer

□ Transport layer:

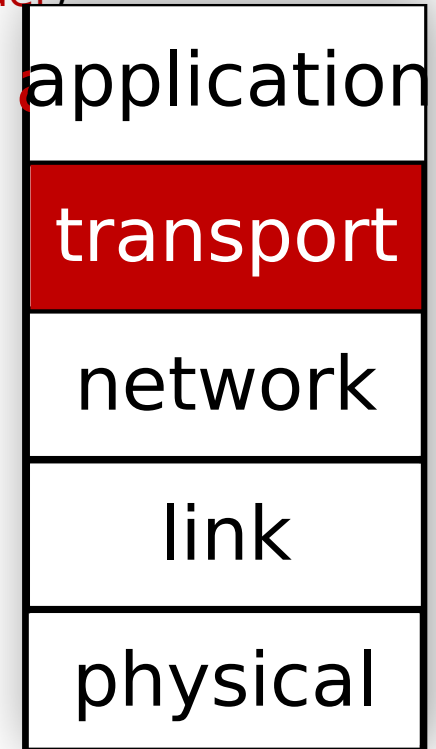
- Provide **process-to-process** delivery
- **TCP**: Connection-oriented, reliable service (**No error, In order**)
- **UDP**: Connectionless, unreliable service



(a) TCP Header



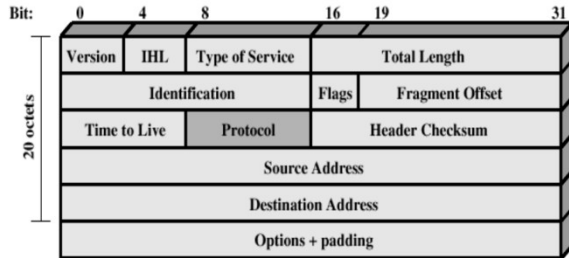
(b) UDP Header



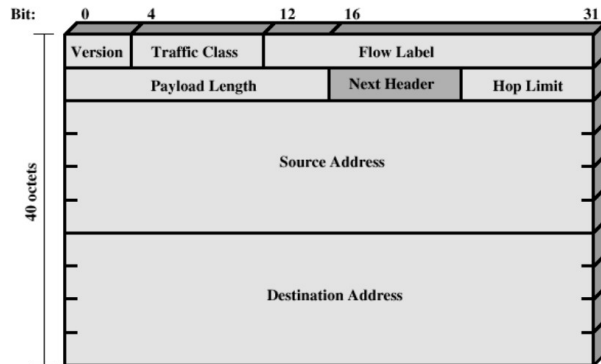
Network Layer

□ Network layer:

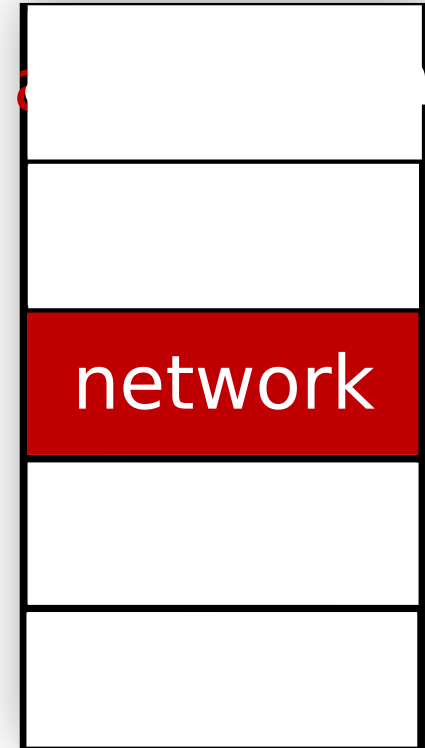
- Provide **Routing** function across multiple networks
- **Uses Internet Protocol (IP) to provide routing functions**
- May provide QoS, congestion control etc
- **Relevant protocols:** ICMP, OSPF, RSVP



- **IPv4 header:**
 - 32 bit source and destination address



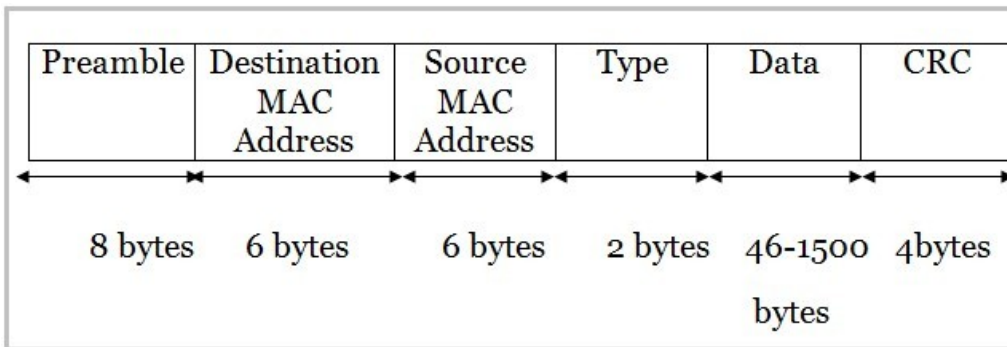
- **IPv6 header:**
 - 128 bit source and destination address



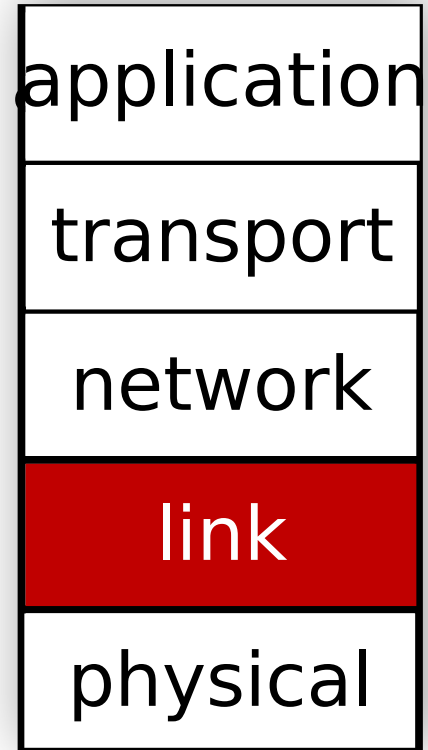
Data Link Layer

□ Data Link layer:

- Transmission of data over the **link** to which the device is attached
- Provide reliable delivery over a **link**
- Flow control and error control
- Sometimes called:
 - Network Access Layer
 - MAC Layer
 - Link Layer
 - Hardware Layer
- **Relevant protocols:** Ethernet, WiFi



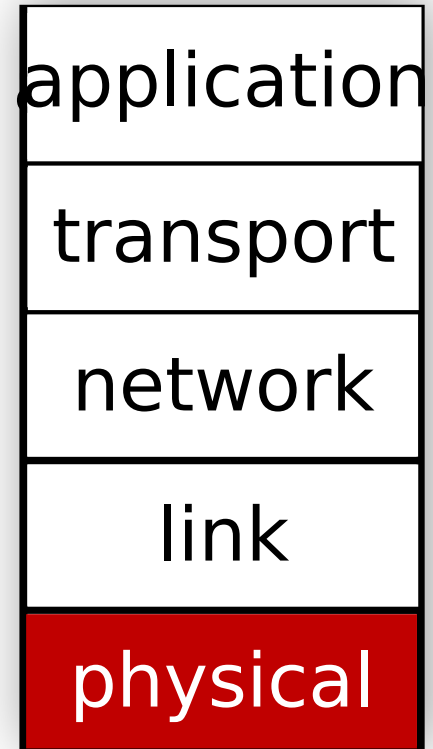
Src: <https://www.minitool.com/lib/ethernet-frame.html>



Physical Layer

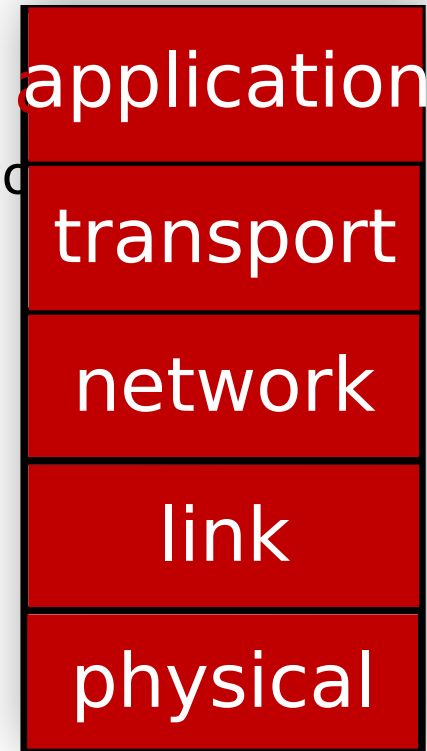
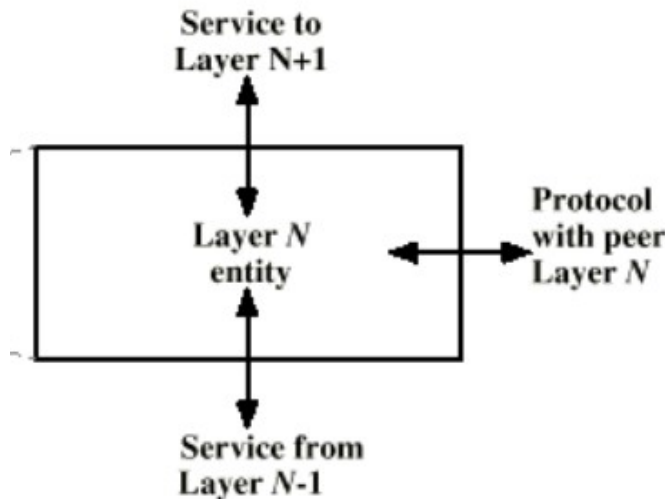
□ Physical layer:

- Transmission of bitstream
- Apply appropriate encoding, modulation techniques

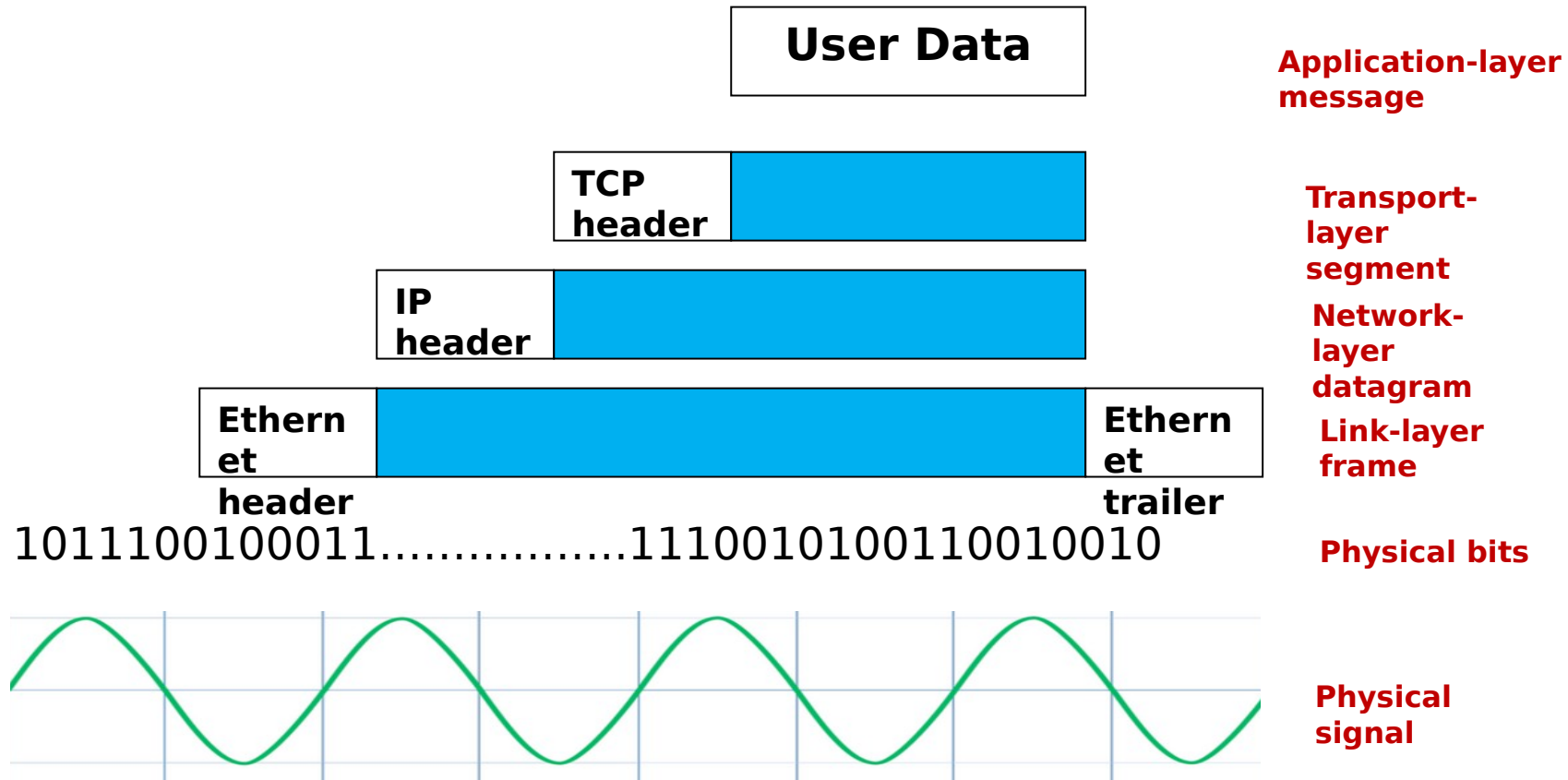


TCP/IP Layers

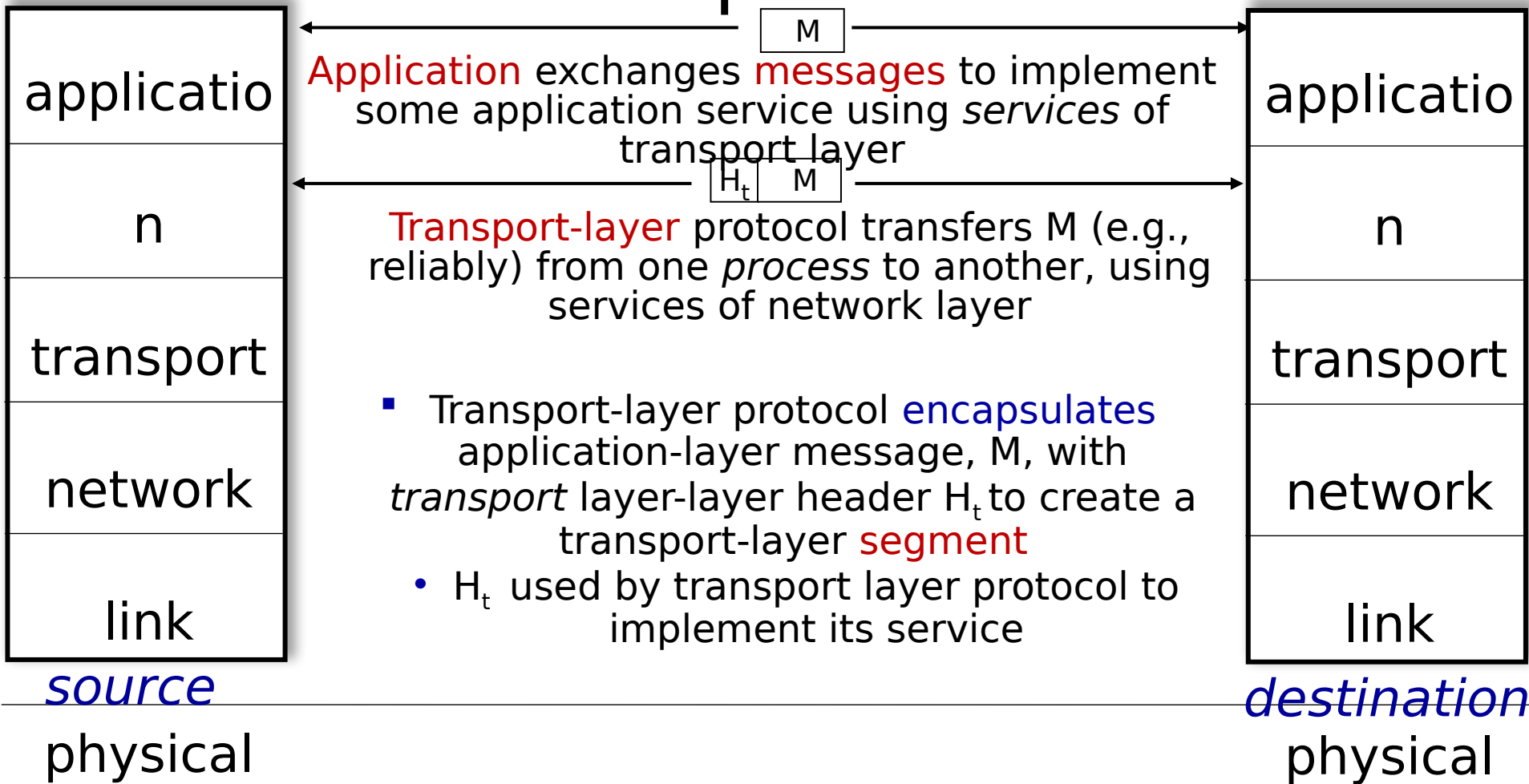
- Layers are arranged in vertical tasks
 - Layer N uses service of layer N-1
 - Layer N provides service to layer N+1
- Peer layers communicate with a protocol



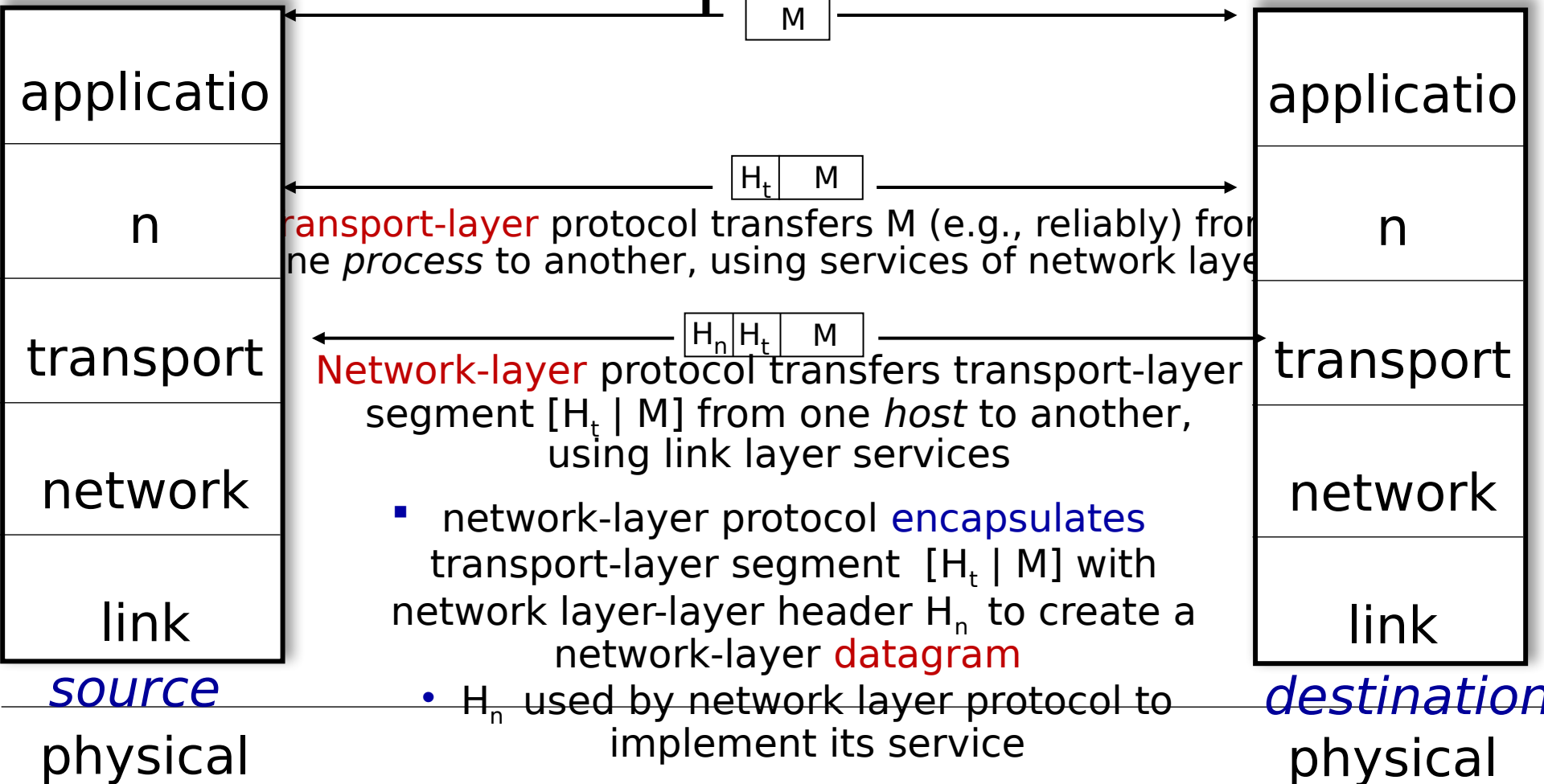
Protocol Data Unit



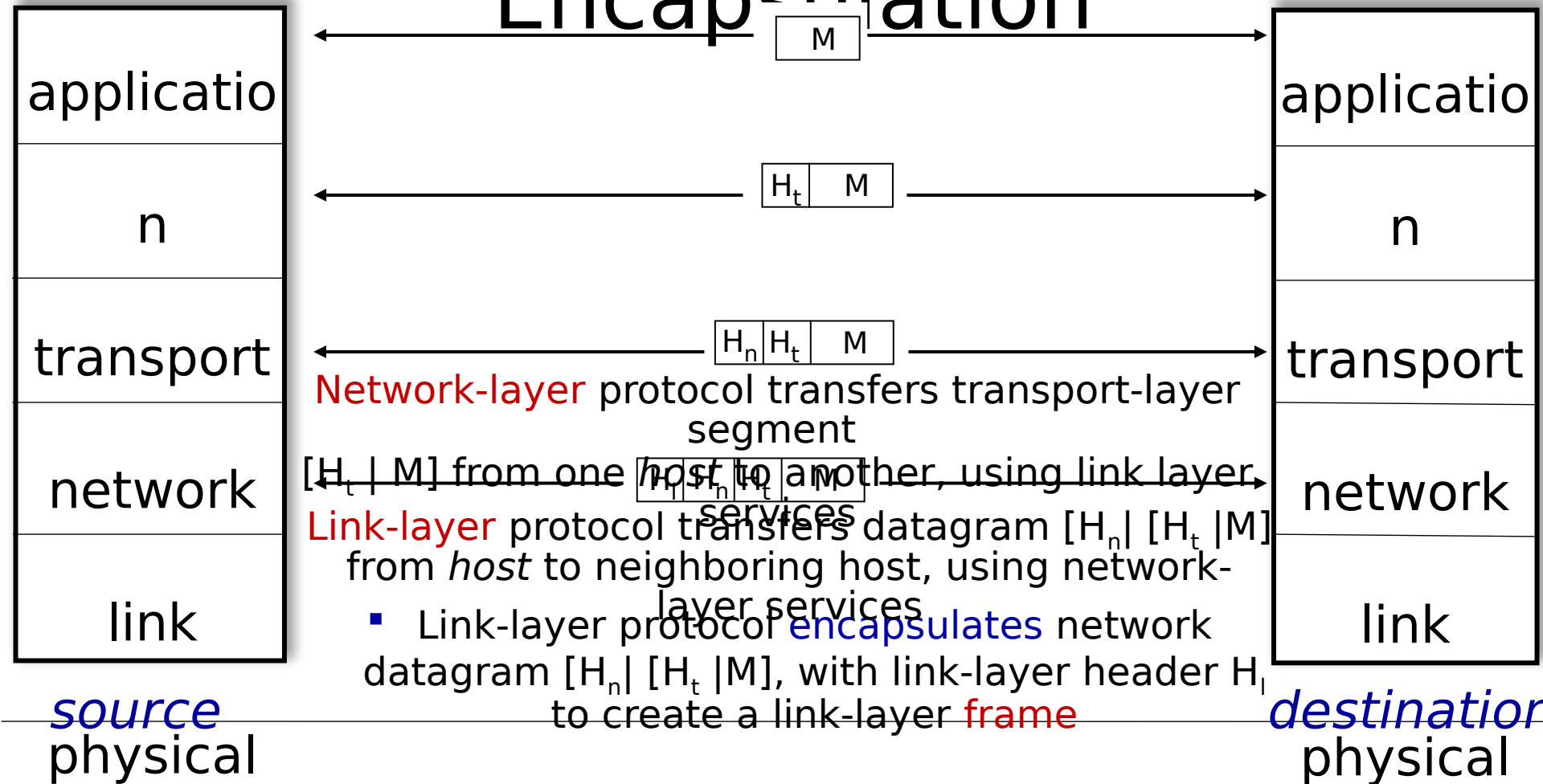
Services, Layering and Encapsulation



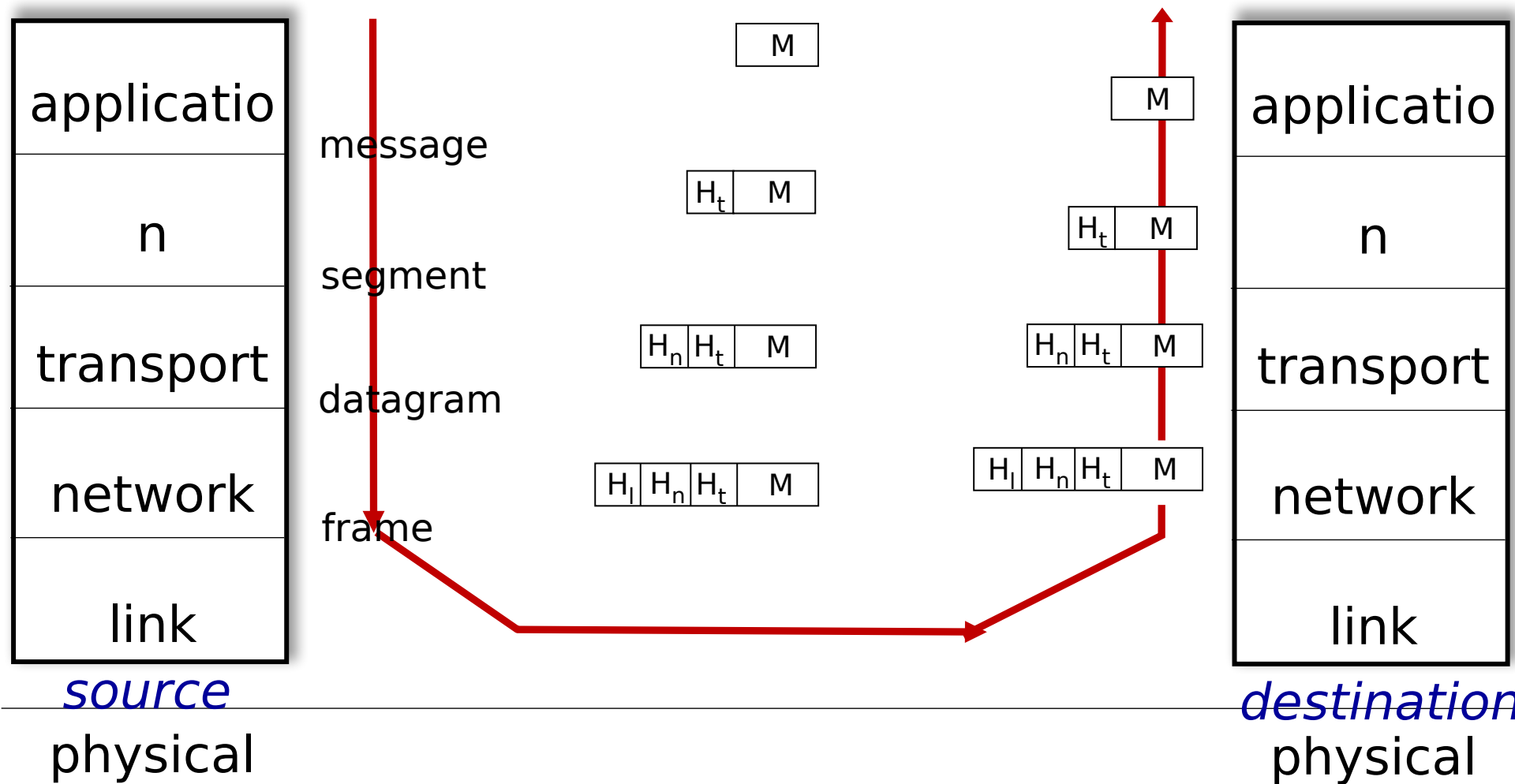
Services, Layering and Encapsulation



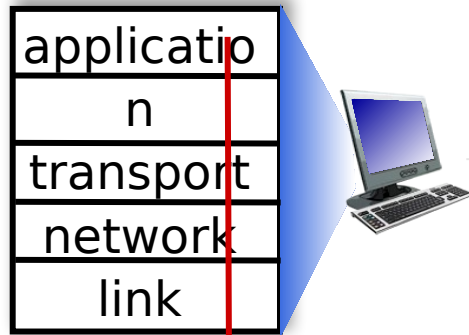
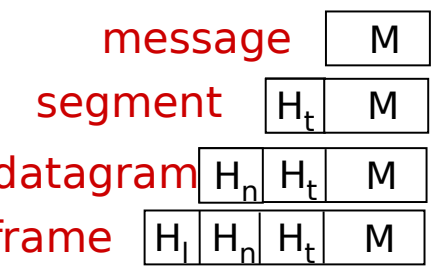
Services, Layering and Encapsulation



Services, Layering and Encapsulation



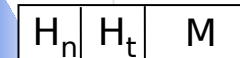
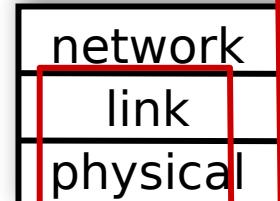
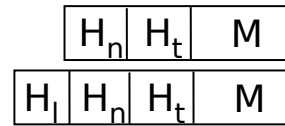
Encapsulation: an end-end view



source

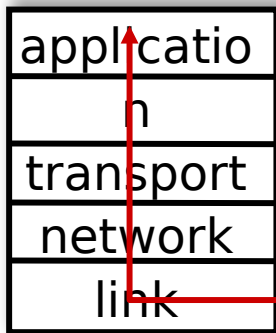
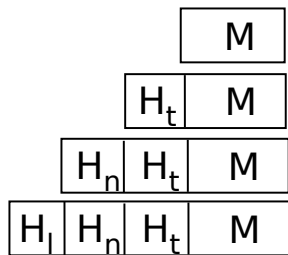


switch



router

destination



physical